Cagayan De Oro College– PHINMA Education Network

JAVA-BASED AQUARIUM SHOP MAINTENANCE AND SCHEDULING SYSTEM FOR AQUARIST

Computer Programming II Project

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1.1 Introduction

As the digital space continues to transform, to prevent life-threatening oversights in an aquarium shop, a structured digital approach is essential. In a business like an aquarium shop, managing and scheduling is crucial. According to Karydis (2020), effective maintenance and scheduling in an aquarium shop are critical for ensuring livestock health, preventing algae, outbreaks and maintaining a professional, appealing display for customers. A structured routine covers daily tasks, balancing customer service with aquatic care. However, when it comes to feeding fish it's also important that providing your fish with the right feeding schedule will ensure growth, disease resistance, vibrant colors, and long, healthy lives. Prasetiawan (2023). For staff, particularly in business, it's very crucial for them to properly perform their tasks by having proper maintenance and scheduling, so they do not forget what they need to do.

As reported by the owner of Aquarist during the initial interview, they`ve been experiencing difficulties in managing their shop when it comes to maintaining the cleanliness of their aquarium. Inadequate aquarium maintenance can negatively affect the health of the fish and may result in their death. Sometimes, due to having so many tasks, they forget that they have filled the aquarium with water, which can cause overflow. Also, the staff need to focus not just on one aquarium, but on many. During busy days, it is unavoidable that feeding the fish is sometimes forgotten, especially when many customers arrive and need to be

attended to, causing them to lose track of the time for feeding. These problems can cause customer satisfaction and business downfall.

The proposed Java-Based Aquarium Shop Maintenance and Scheduling System for Aquarist aims to improve the processes to make aquarium tasks faster and easier through a console-based program. The staff can log in, register and check tasks or activities given. At the same time, the admin can give the task to do, delete, schedule and review all the checked tasks. This process can help to assist the staff in managing and taking care of the aquarium , prevent the fish from dying, and make their daily work easier and faster. Through the development & creating a system that can plan and organize aquarium maintenance tasks effectively this allows to improve that important activities such as cleaning tanks, checking the water, and inspecting the equipment are done on time and in the correct way.

1.2 Statement of the Problem

General Problem

Aquarist confronts a challenge because of lack and does not have a proper system to manage and monitor maintenance, scheduling, and feeding tasks, which leads to forgotten or delayed activities that negatively affect fish health and overall business operations.

Specific Problem

- **Difficulty in Managing Aquarium Maintenance.**

Because usually the staff forgot about the aquarium maintenance, which may cause water overflow and poor aquarium maintenance that can negatively affect the health of the fish and may lead to fish deaths.

- **Lack of Aquarium Maintenance Scheduling.**

The aquarium maintenance tasks in shops are sometimes forgotten or not done in time because there are no existing systems to monitor maintained activities. This may lead to poor aquarium condition or water overflow, which can also affect the appearance of the shop and reduce customer satisfaction, which may affect the overall impact of the business.

- **Lack of Fish Feeding Schedule Monitoring.**

Feeding the fish is an important task in aquarium shops. However, the business has no existing system to schedule or remind tasks for feeding time, which may result in missed or not done regularly.

1.3 General Objectives

The main goal of this study is to develop a Java-based Maintenance and Scheduling System that can help manage, organize, and monitor aquarium maintenance and fish feeding tasks efficiently.

1.4 Specific Objectives

- To help staff manage and maintain the aquarium, avoid fish deaths, and make their work easier and faster.
- To develop a system that can effectively schedule aquarium maintenance tasks, ensuring that all necessary activities such as cleaning, water checking, and equipment inspection are properly planned and carried out on time.
- To help staff by creating a feature that provides timely reminders regarding maintenance activities and fish feeding schedules, in order to prevent missed or delayed tasks.

Scope & Limitation

1.5 Scope

The proposed Java-Based Aquarium Shop Maintenance and Scheduling System focuses on providing a simple way for staff to remember their daily tasks in maintaining the aquarium and to check the schedule for feeding the fish.

- The system allows the admin to create and manage maintenance and feeding tasks to ensure that all aquariums are properly maintained.
- The admin can input, update, and delete task details such as assistant name, type of task, and date and time of the activity.
- The admin can mark tasks as completed once they are finished by the assistant and can log out from the admin portal..
- The assistant can view assigned tasks, mark them as completed after finishing their work, and log out from the assistant portal.
- The system helps monitor daily tasks such as aquarium cleaning and fish feeding to avoid missed or delayed activities that may affect fish health.
- The system provides login access for both admin and assistant to control and manage system functions.

1.6 Limitation

This study is limited to the development of a Java-based Maintenance and Scheduling System designed specifically for managing aquarium maintenance and fish feeding tasks within a single aquarium shop. The system focuses only on scheduling, reminders, and task monitoring. It will be used only by the shop owner or staff and is not accessible to customers or external users. Additionally, the system depends on manual input of data and schedules, which means its effectiveness relies on the accuracy and consistency

of the user. The system runs exclusively on a console and supports only a single user at a time. It does not include advanced features like file uploads, notifications, or data encryption, making it suitable solely for demonstrating the basic reminder maintenance and scheduling tasks in an offline setup.

1.7 System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent.
- RAM: At least 2 GB.
- Storage: Minimum 100 MB of free disk space.
- Input: Keyboard for commands.
- .Monitor: Any display that supports text output (console view)
- Network: None required.

Software Requirements

- Operating System: Windows or macOS
- Programming Language: Java
- Code Editor: IDE NetBeans
- Environment: Command-line interface.
- Console or Terminal: Console

User- Staff/Assistant

1. **Register or Log In**

   Allow the user to create an account or login.

2. **View Task**

   The assistant can view assigned tasks.

3. **Mark as Completed**

   The assistant can mark them as completed after finishing the task.

4. **Log Out**

   Allow to log out from the assistant portal.

User- Admin/Owner

1. **Log In**

   Admin can securely log in using password.

2. **Create Task**

   Allows the admin to create and manage maintenance and feeding tasks.

3. **Update Task**

   The admin can input, update and delete task details such as assistant name, type of task and date and time of the activity

4. **Delete Task**

   Admin can delete tasks once they complete and can replace it with other tasks.

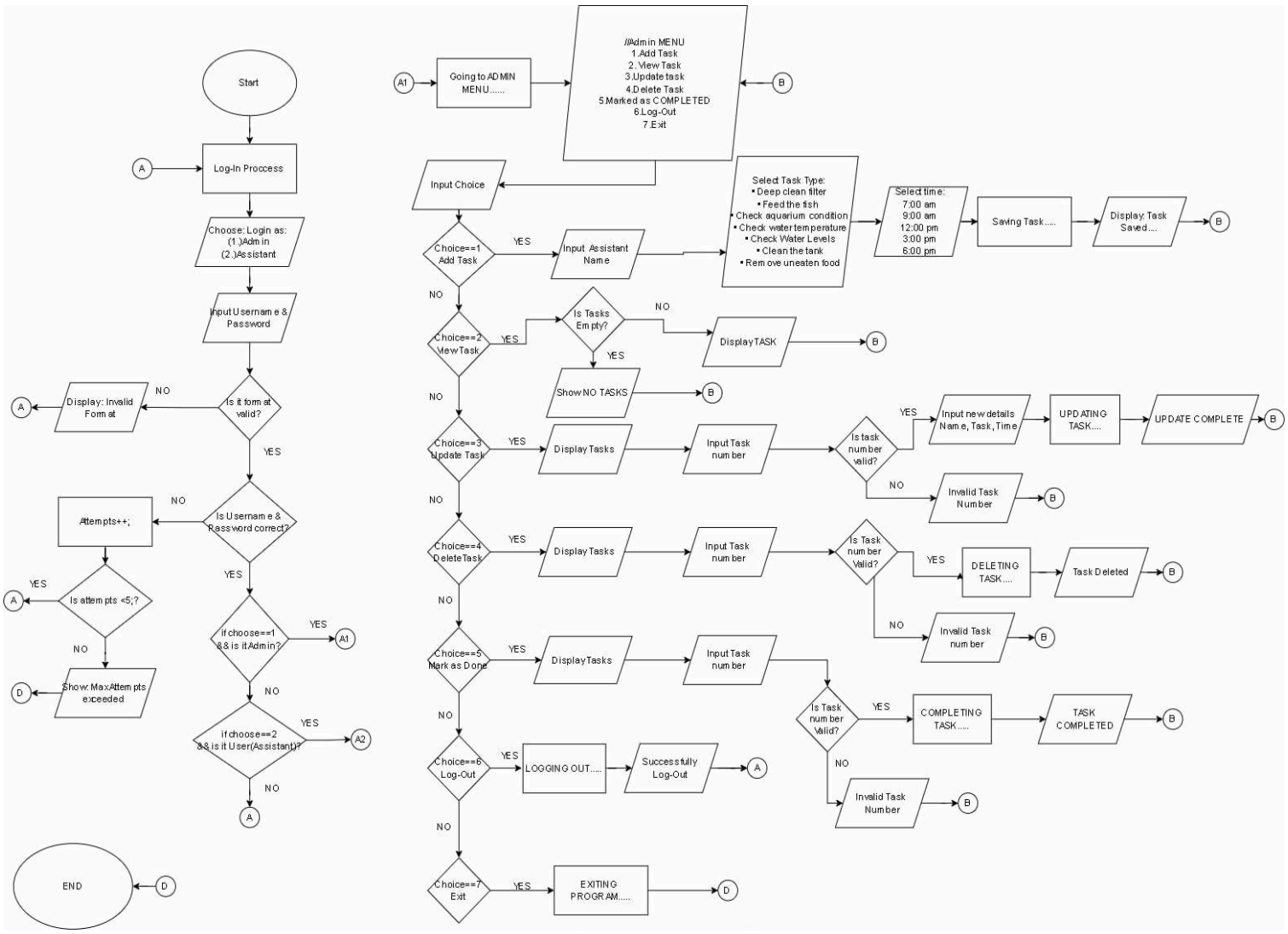
5. **Mark Task Completed**

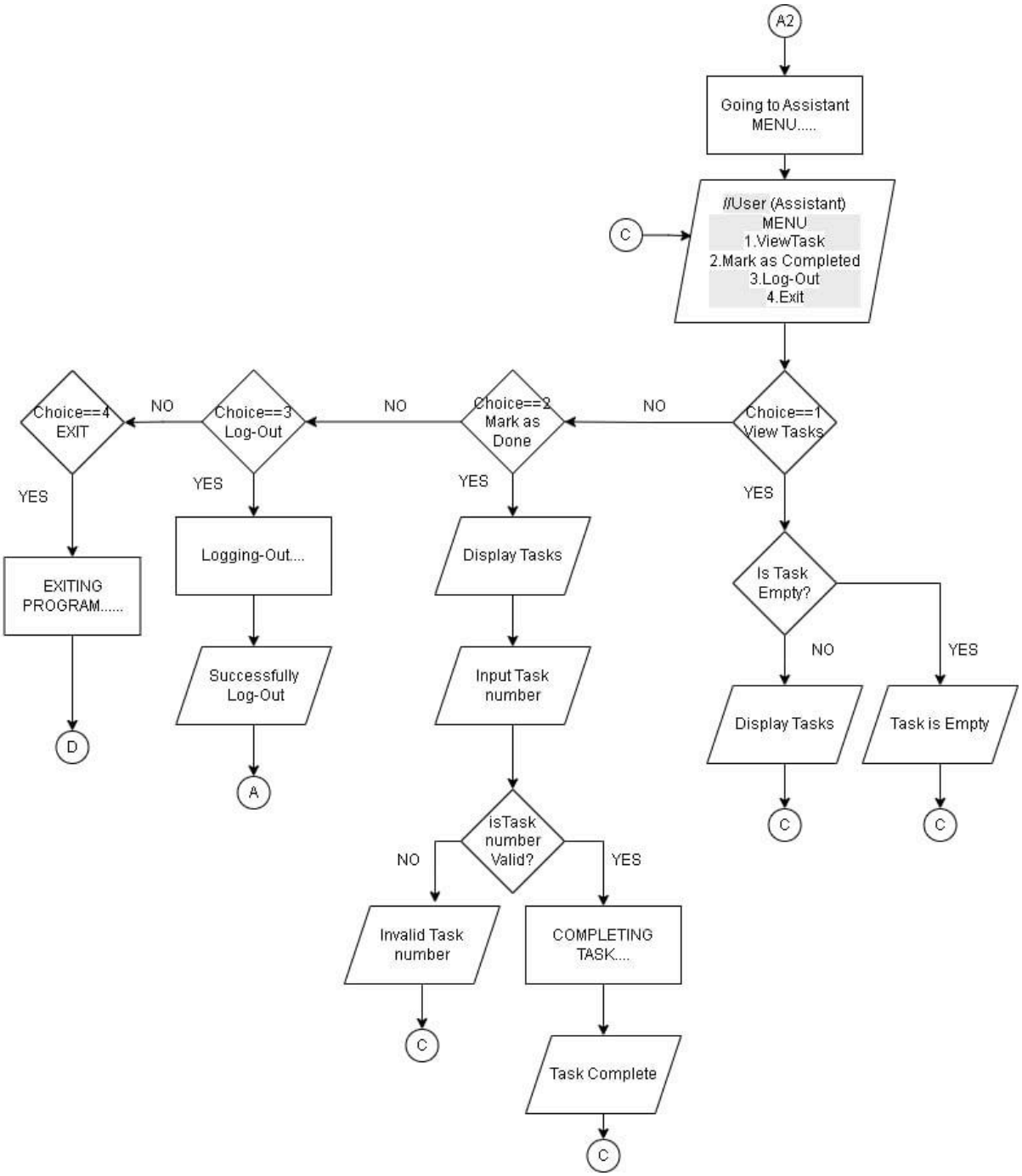
   The admin can mark tasks as completed once they are finished by the assistant.

6. **Log Out**

   Admin can log out from the admin portal.

1.8 Flowchart




A2
Going to Assistant MENU.....
C
//User (Assistant)
MENU
1.ViewTask
2.Mark as Completed
3.Log-Out
4.Exit
Choice==1 View Tasks
NO
YES
Choice==2 Mark as Done
NO
YES
Choice==3 Log-Out
NO
YES
Choice==4 EXIT
YES
Is Task Empty?
NO
YES
Display Tasks
Task is Empty
C
C
Display Tasks
Input Task number
isTask number Valid?
NO
YES
Invalid Task number
COMPLETING TASK....
C
Task Complete
C
Logging-Out....
Successfully Log-Out
A
EXITING PROGRAM.....
D

The flowchart shows a simple task management system. It starts when the user logs in by entering a username and password. The system checks if the input is correct, the system checks the type of user.

If the user is an admin, they can add, view, update, delete, or complete tasks. They can also log out or exit the system. If the user is a regular user or assistant, they can only view tasks, mark tasks as completed, log out or exit. When doing actions like updating or completing a task, the system asks for a task number and checks if it is valid.

If the input is wrong, the system shows an error message. If everything is correct, the action is completed. The process continues until the user chooses to log out or exit, which ends the system.

1.9 Code Snippet of Java-Based Aquarium Shop Maintenance and Scheduling System

```
ckage com.mycompany.java.based.aquarium.shop.maintenance.and.scheduling.system;
port java.util.Scanner;
port java.util.ArrayList;
port java.time.LocalTime;
port java.time.LocalDate;
port java.time.format.DateTimeFormatter;
blic class JavaBasedAquariumShopMaintenanceAndSchedulingSystem {
  static class Task {
    String name , type,date, status;
    Task (String name, String type, String date){
        this.name = name;
        this.type = type;
        this.date = date;
        this.status = "pending";
    }
  }
  static ArrayList<Task> Tasks = new ArrayList<>();

  public static void main(String[] args) {

      Scanner input = new Scanner(System.in);
      //admin
     String adminusername ="admin123";
      String adminuserpass ="Admin123";
     //user
     String userUsername ="User1234";
      String userUserpass ="User=321";
       //Regex pattern
       String usernamepattern ="^[a-zA-Z0-9]{8,12}$";
       String  passwordpattern ="^(?=.*[A-Z])(?=.*[a-z])(?=.*\\d).{8,12}$";
       //attempts
       int attempts = 1;
       while(true){
       boolean isAdmin = false;
       boolean islogin = false;
```

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